

CSE 4621: Machine Learning

Cluster Analysis

Lecture 11

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Lecture Outline

- Unsupervised Learning
- Cluster Analysis
 - Usage of clustering
 - Quality of a good cluster
 - Considerations in clustering
- Overview of Clustering Methods
- K-means clustering
- K-medoids clustering
- Choosing the value of k
- Supervised clustering (KNN)

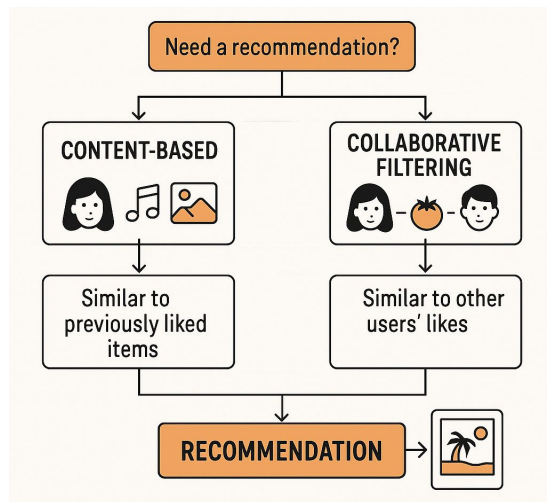


Unsupervised Learning

Find undetected patterns in a dataset with no pre-existing labels and with minimum human supervision.

Suppose you want an algorithm that can recommend reels based on your activities

- Cluster Analysis
- PCA



What is Cluster Analysis

An alien invasion is imminent and only you can stop it!



What is Cluster Analysis

Cluster: A collection of data objects

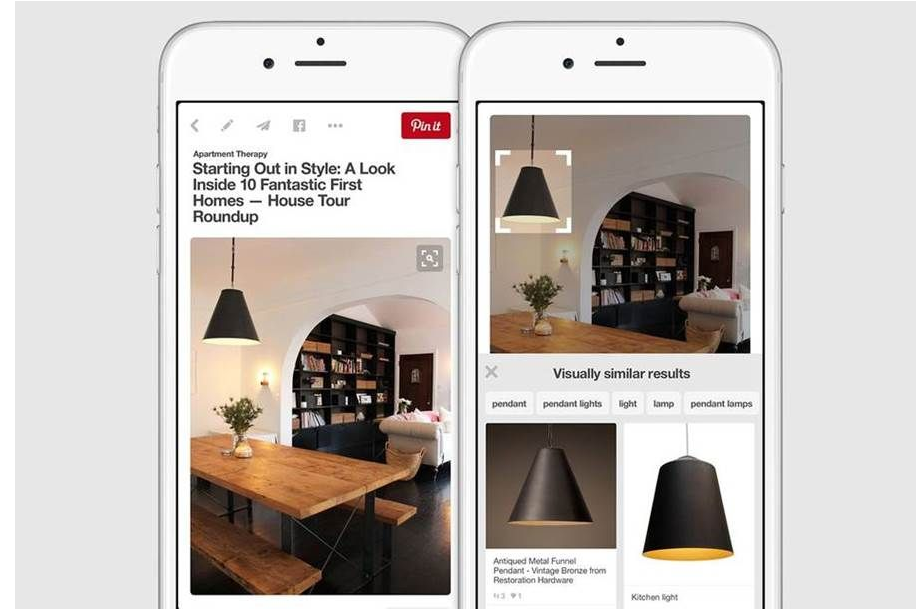
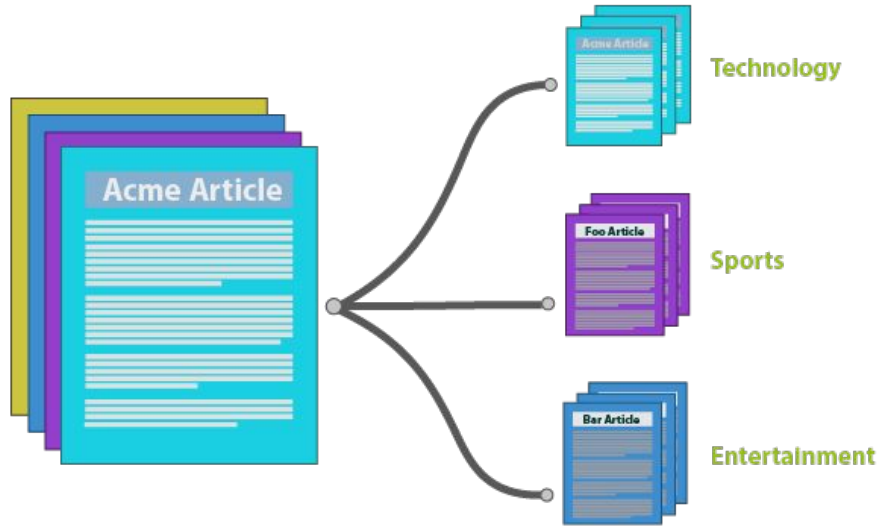
- similar (or related) to one another within the same group
- dissimilar (or unrelated) to the objects in other groups

Cluster Analysis is the study of finding similarities between data points according to the characteristics found in the data and grouping similar data objects into clusters

No predefined classes → Learning from experience

- Stand-alone data analysis tool
- Pre-processing step for another algorithms

Applications of Clustering



Applications of Clustering

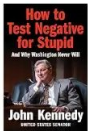
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 Propel Electrolyte Enhanced Water Sports Drink, Grape Flavored Water, Zero Calorie & Zero Sugar Hydration with... ★★★★★ 42,834	 Core Power Protein Shake, Chocolate, 26g Bottle, 14oz, 12 Pack ★★★★★ 218,946 \$19.99	 Amazon Basics Slim, Velvet, Non-Slip Suit Clothes Hangers, Pack... ★★★★★ 218,946 \$19.99	 Core Power Elite High Protein Shake, Chocolate, 42g Bottle, 14oz, 12 Pack ★★★★★ 37,509	 Starbucks K-Cup Coffee Pods, Medium Roast Coffee, Pike Place Roast for Keurig Brewers, 100% Arabica, 4 boxes (96 pods total) ★★★★★ 37,509	 PUR Gum Aspartame Free Chewing Gum 100% Xylitol Natural Spearmint Flavored Gum, 55 Pieces (Pack of 3) ★★★★★ 37,509	 Nespresso Capsules Vertuo, Barista Flavored Pack, Medium Roast Coffee, 30 Count Coffee Pods, Brews 7.8oz. ★★★★★ 37,509

Applications of Clustering



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Top 34 documents retrieved for the query forensic psychiatry

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Links to many online courses including Health and Medicine, Nursing, Science and Technology, Business and MBA, Law and Graduate, and Computers and IT. Visit our website to get your degree and advance your career today. - www.geocities.com/fk567us/shopeducation.html

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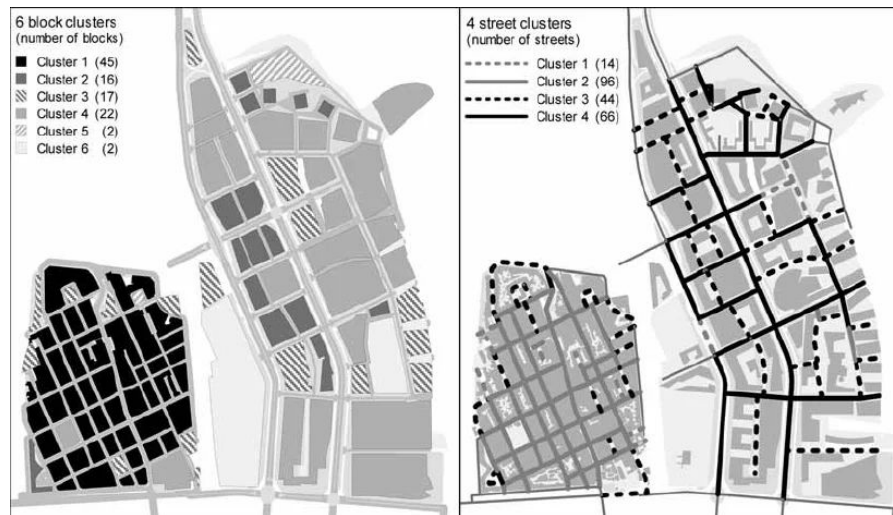
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1. Forensic Psychiatry & Medicine: Managed Health Care and...

... opinion can be effectively communicated by written report, deposition, or courtroom testimony. The applications of forensic psychiatry are widespread in settings ranging from health care and the...

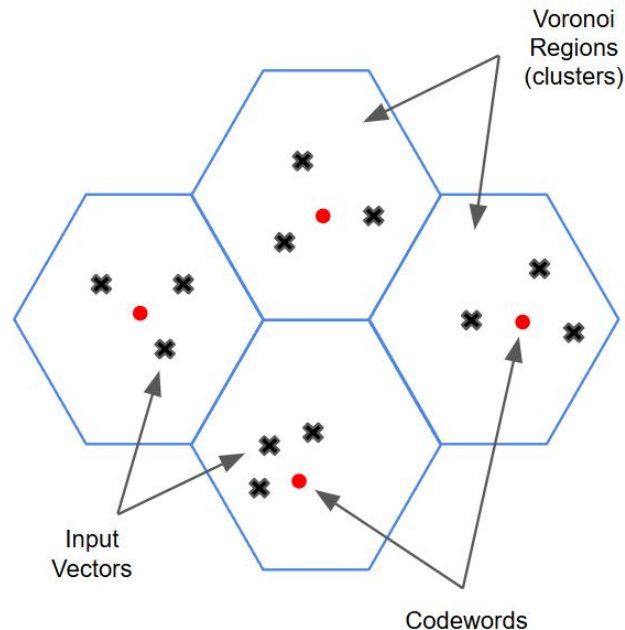
URL: www.forensic-psych.com/

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Clustering as a Pre-Processing Tool

- Summarization → Before regression / classification or PCA
- Compression → Vector quantization
- Localization → Localized search to a few small clusters
- Outlier Detection → Outliers are often far from established clusters



Quality of a Cluster

A good clustering algorithm will produce high quality clusters which have:

- high intra-class similarity: cohesive within clusters
- low inter-class similarity: distinctive between clusters

The quality of a clustering method depends on the *similarity measure* and its ability to discover some or all of the hidden patterns.

Similarity metric: Similarity is expressed in terms of a distance function $d(i, j)$

Weights should be associated with different variables based on applications and data semantics

Considerations for Cluster Analysis

- Partitioning Criteria
 - Single level vs. hierarchical partitioning (often, multi-level hierarchical partitioning is desirable). E.g. Politics, Sports: Football, Cricket, volleyball, etc.
- Separation of Clusters
 - Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)
- Similarity Measure
 - Distance-based (e.g., Euclidian, road network, vector) vs. connectivity-based (e.g., density or contiguity)
- Clustering Space
 - Full space (often when low dimensional) vs. subspaces (often in high-dimensional clustering)

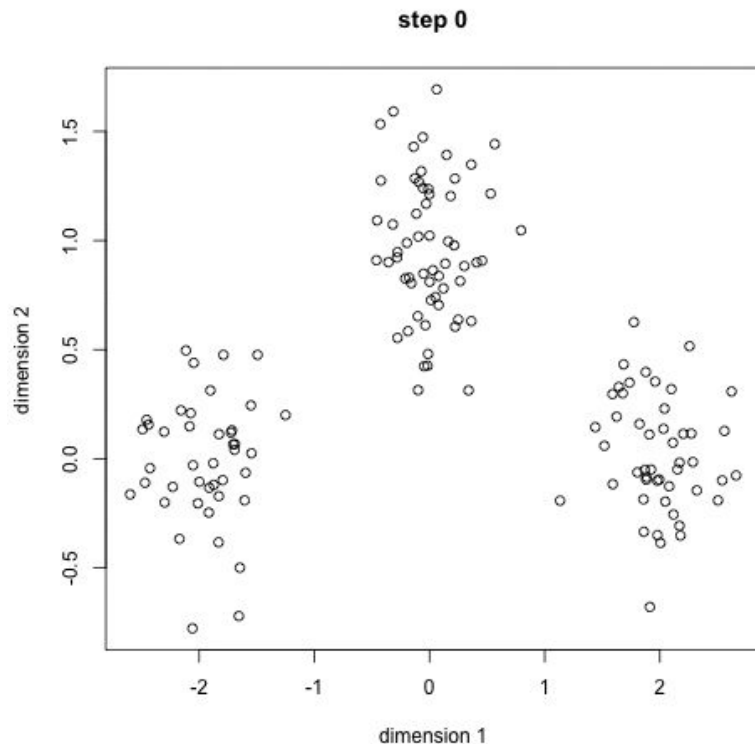
K-Means Clustering

Input: k (number of clusters) , D (dataset of n objects)

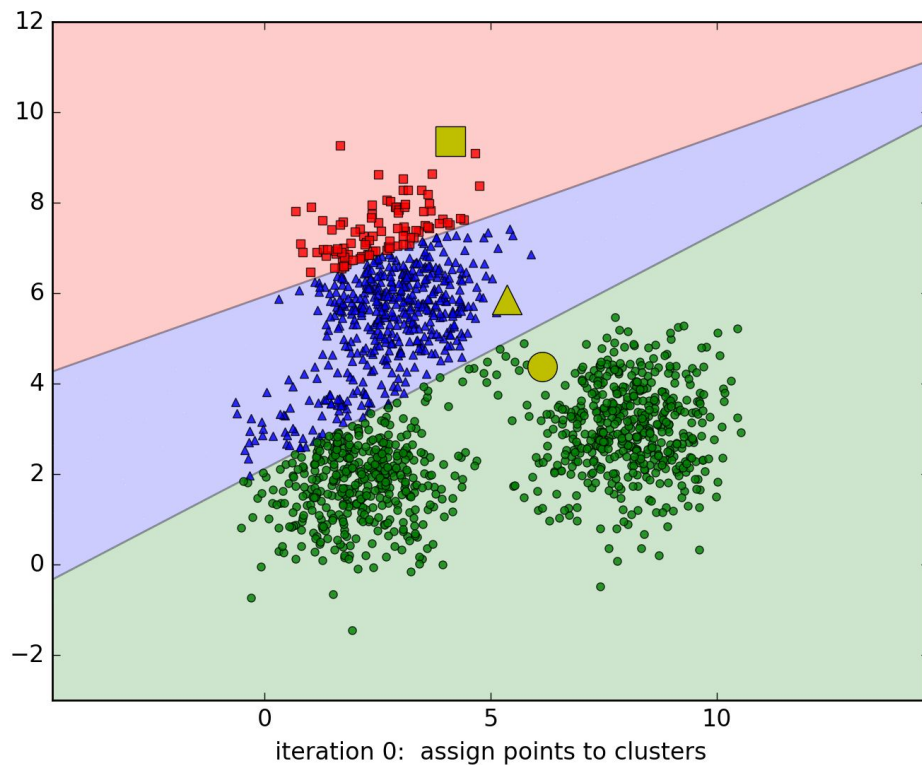
Output: A set of k clusters

Method:

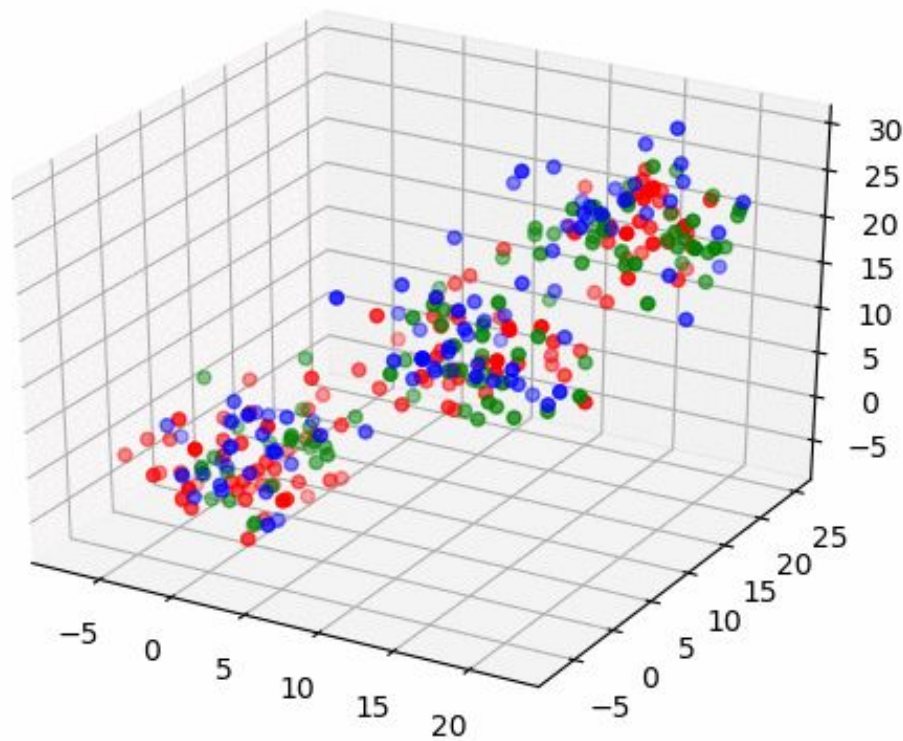
- Randomly choose k objects from D as initial centroids
- Repeat until no change of centroids:
 - Assign each object to the cluster it is most similar to
 - Update the cluster centroids as the mean of the cluster



K-Means Clustering



K-Means Clustering



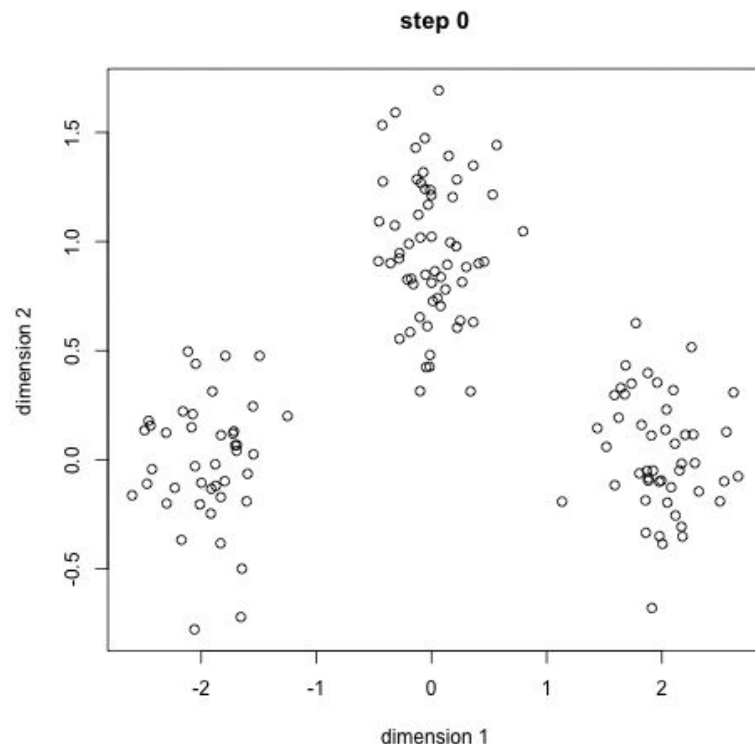
K-Medoids Clustering

Input: k (number of clusters) , D (dataset of n objects)

Output: A set of k clusters

Method:

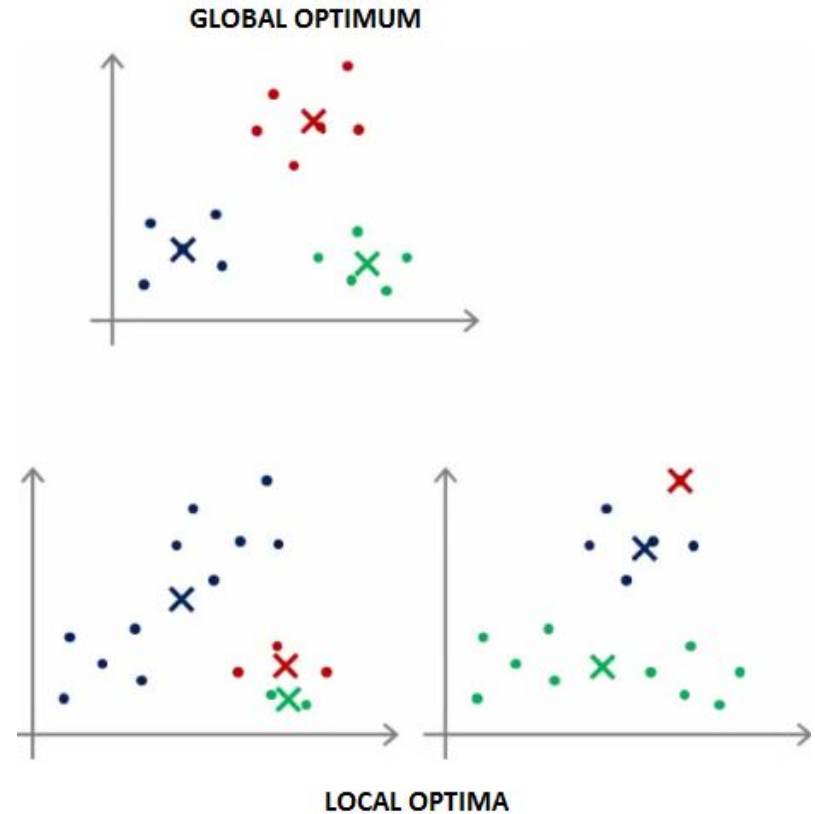
- Randomly choose k objects from D as initial centroids
- Repeat until no change of centroids:
 - Assign each object to the cluster it is most similar to
 - Update the cluster centroids as the median of the cluster



Local Optima

As the centroids are randomly initialized, they can get stuck in a local optima

One naive solution is to repeat the clustering algorithm multiple times and pick the one with the lowest cost $E(c_1, c_2, \dots, c_k)$



Non-convex Clusters

k-means cannot represent density-based clusters

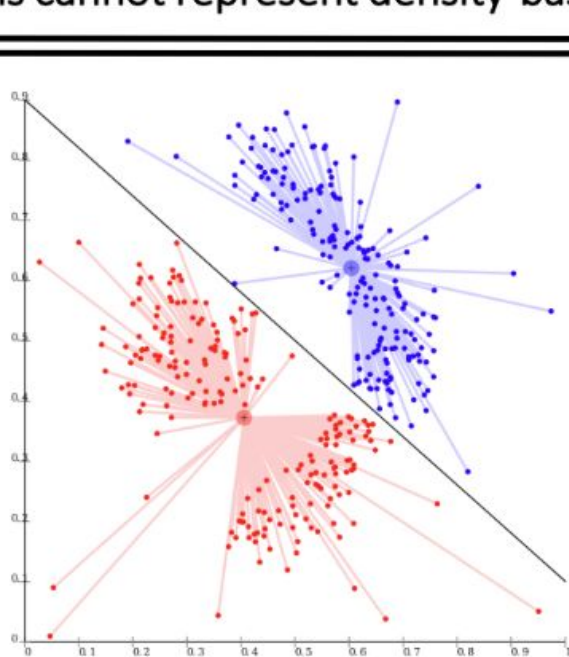


Fig: Convex Shaped

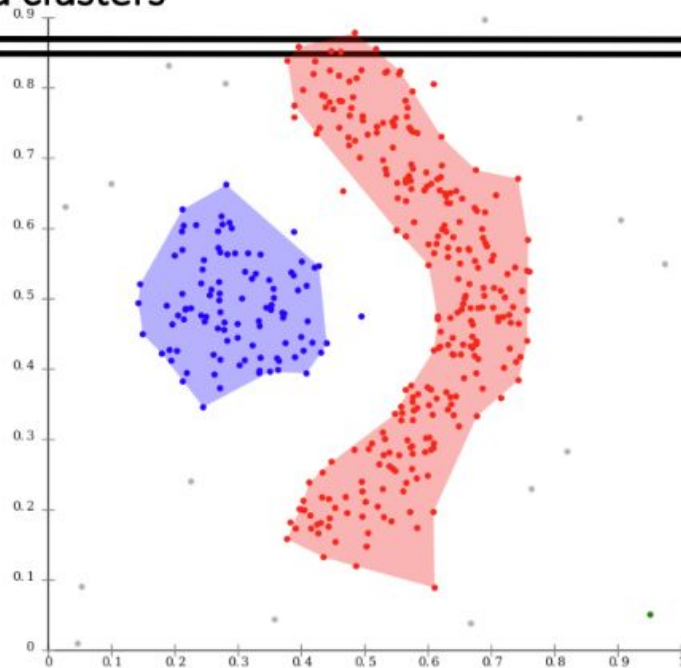
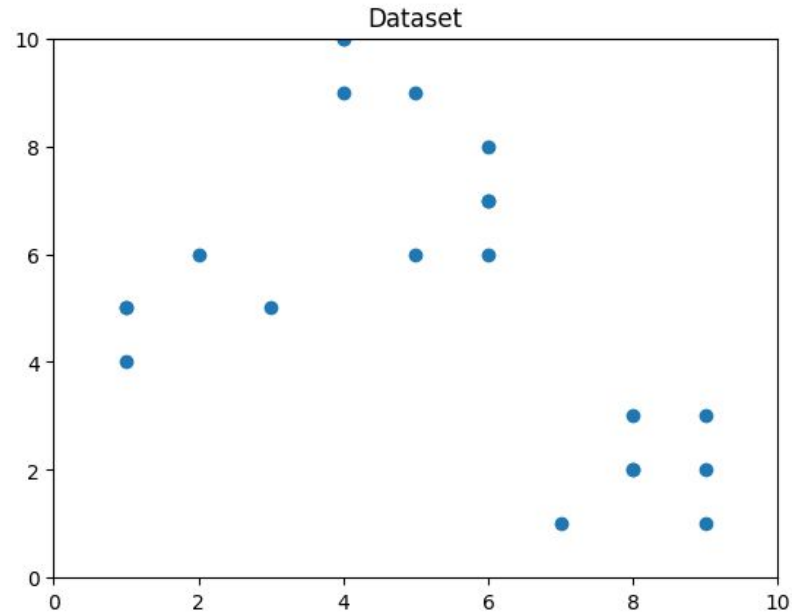


Fig: Non-Convex Shaped

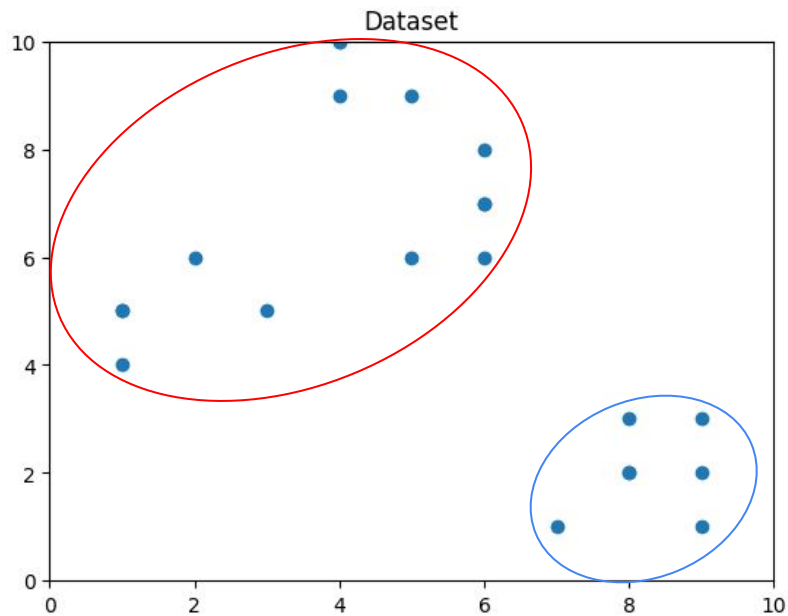
Choosing the number of clusters

How to choose the value of k ?



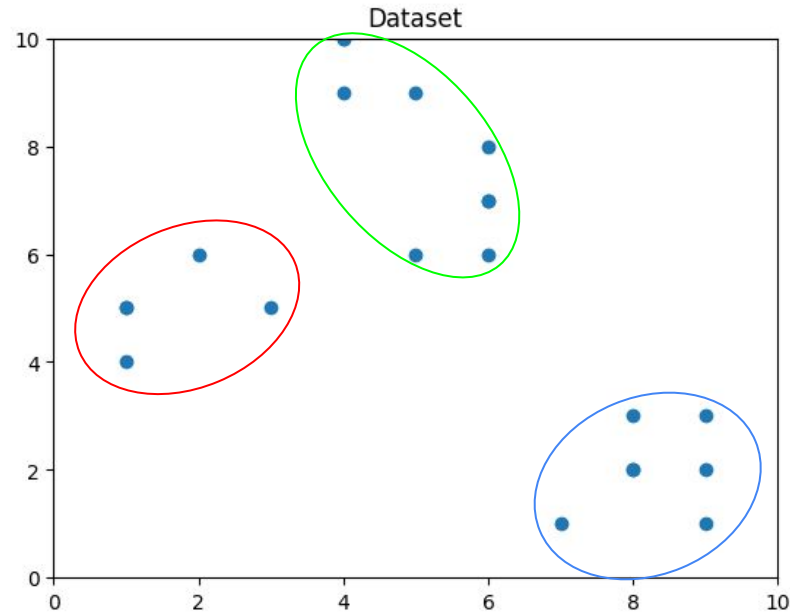
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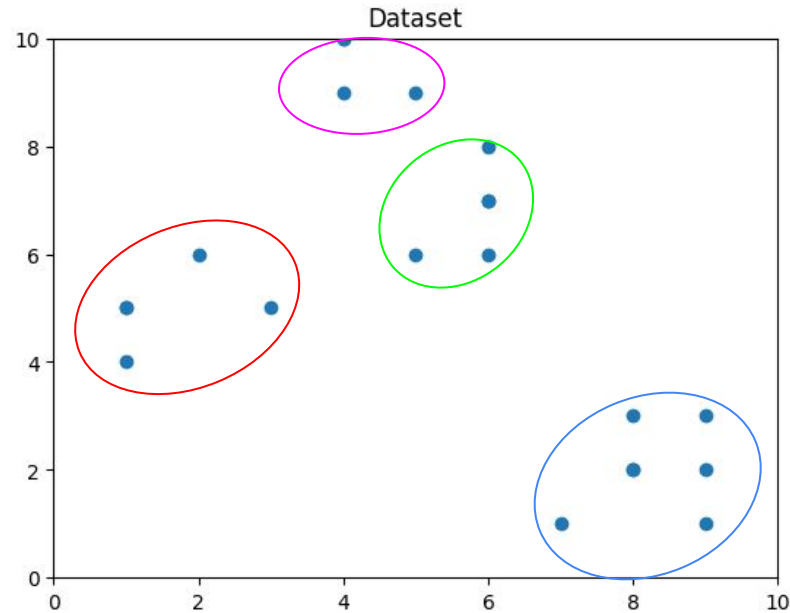
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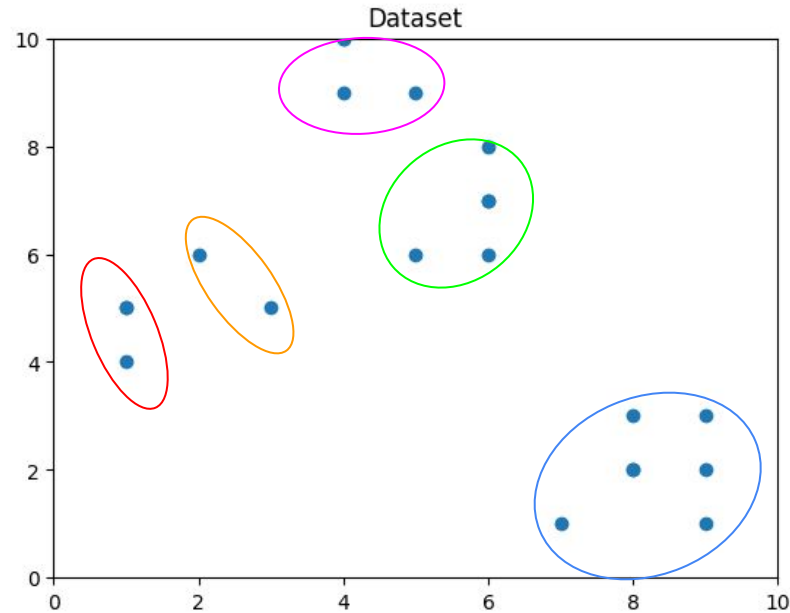
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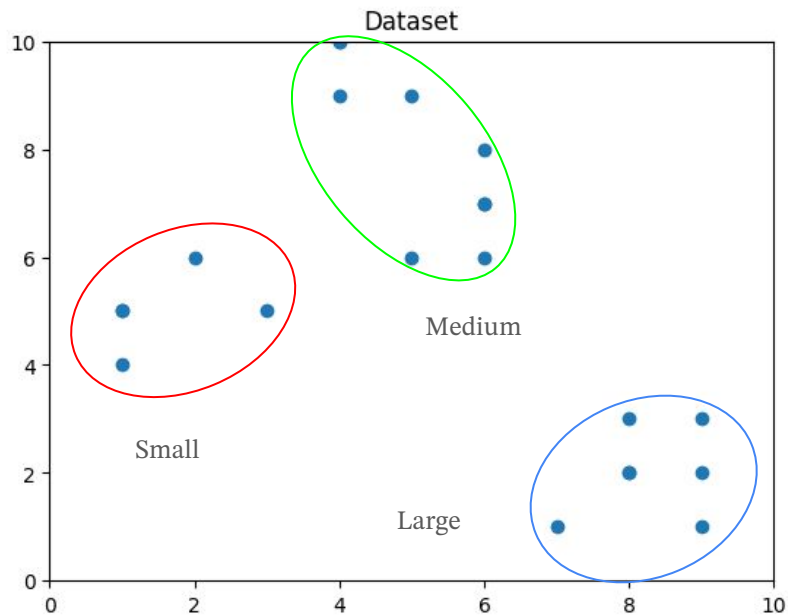
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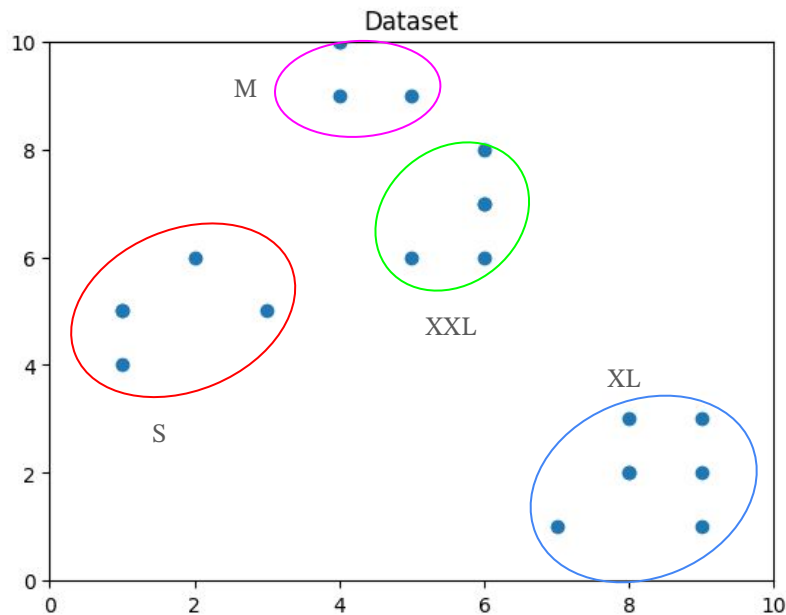
Choosing the number of clusters

The value of k depends on the downstream task



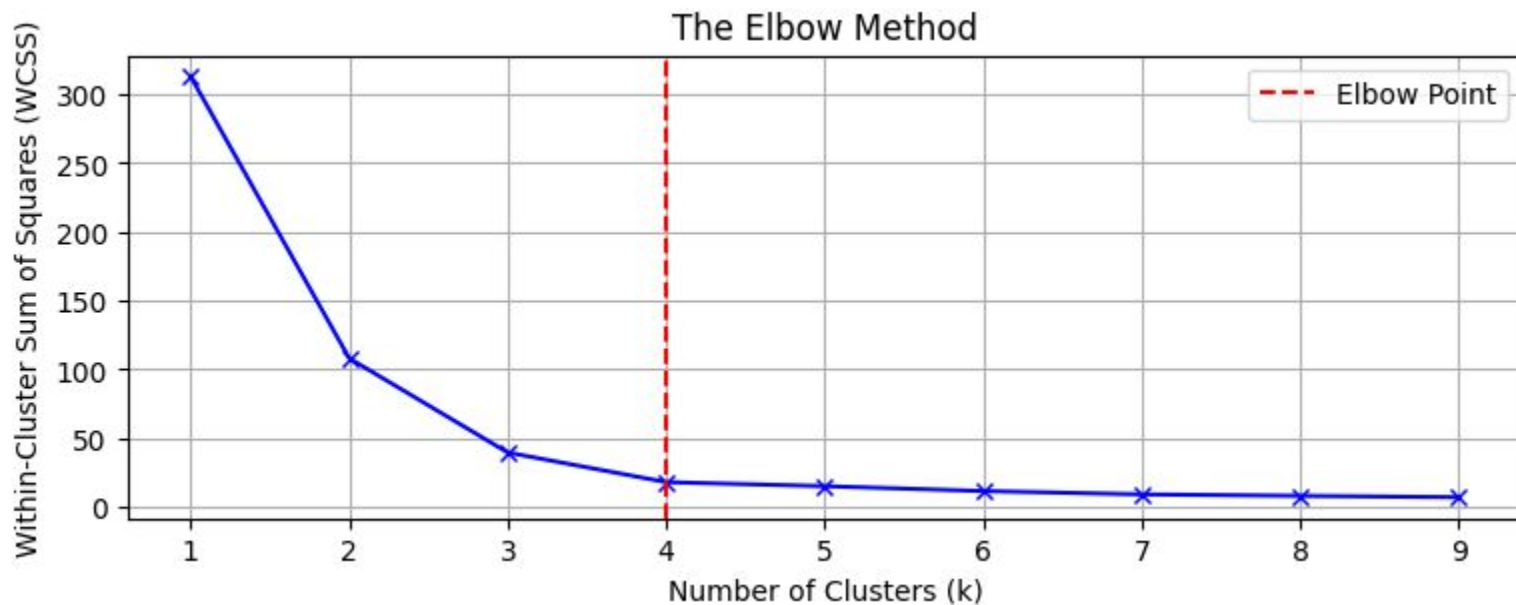
Choosing the number of clusters

The value of k depends on the downstream task



Choosing the number of clusters

Use the elbow method

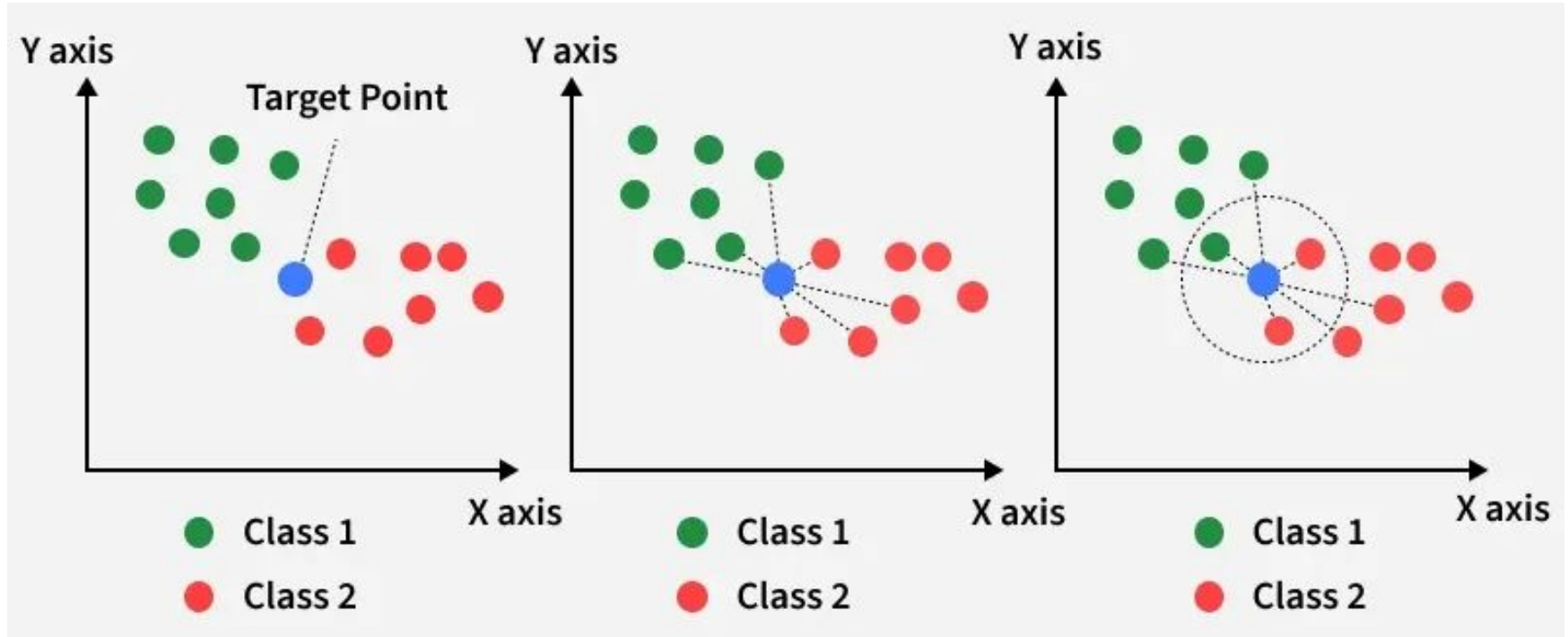


Supervised Clustering

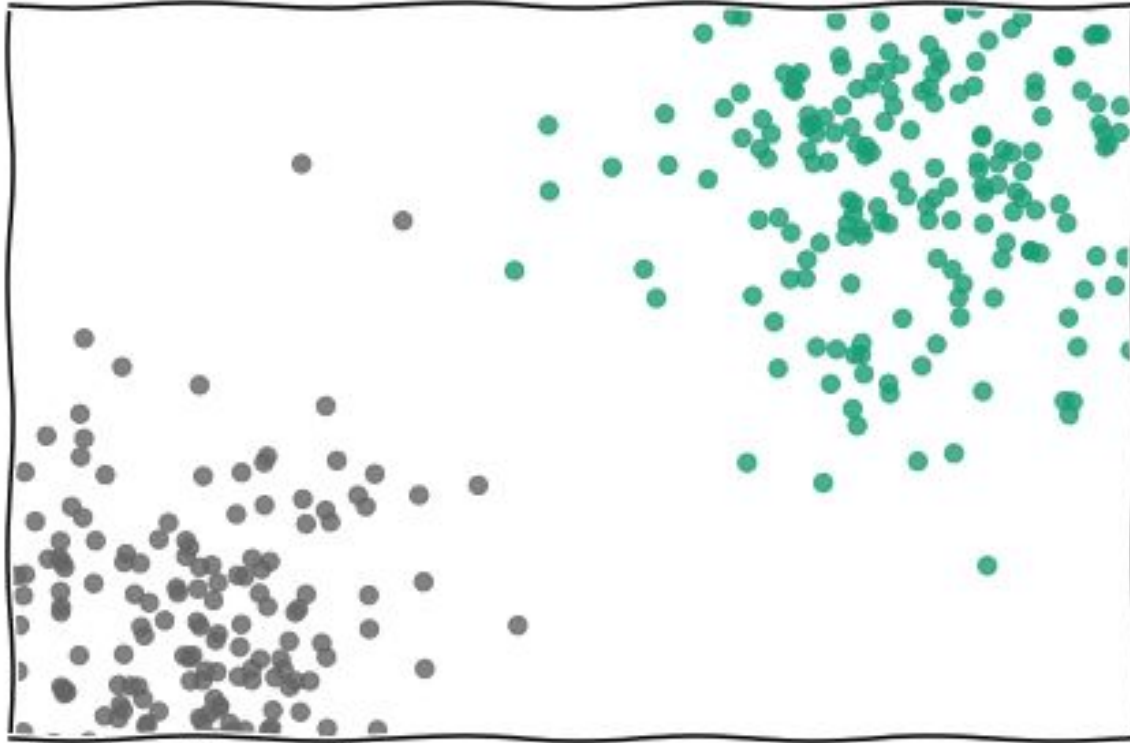
In some cases, clustering analysis can be used in supervised settings. One such example is K-Nearest Neighbors (KNNs)



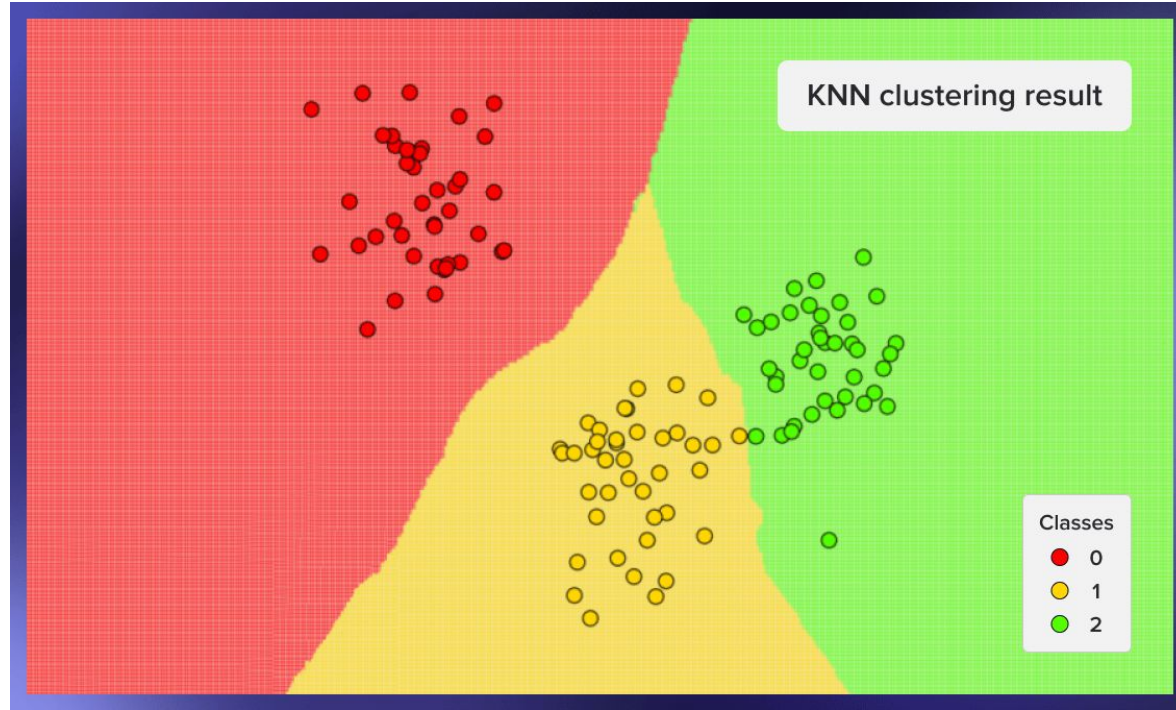
K-Nearest Neighbors



K-Nearest Neighbors



K-Nearest Neighbors



Recommended Resources

1. [Pinterest Sharpens its Visual-Search Skills](#)

Thank you